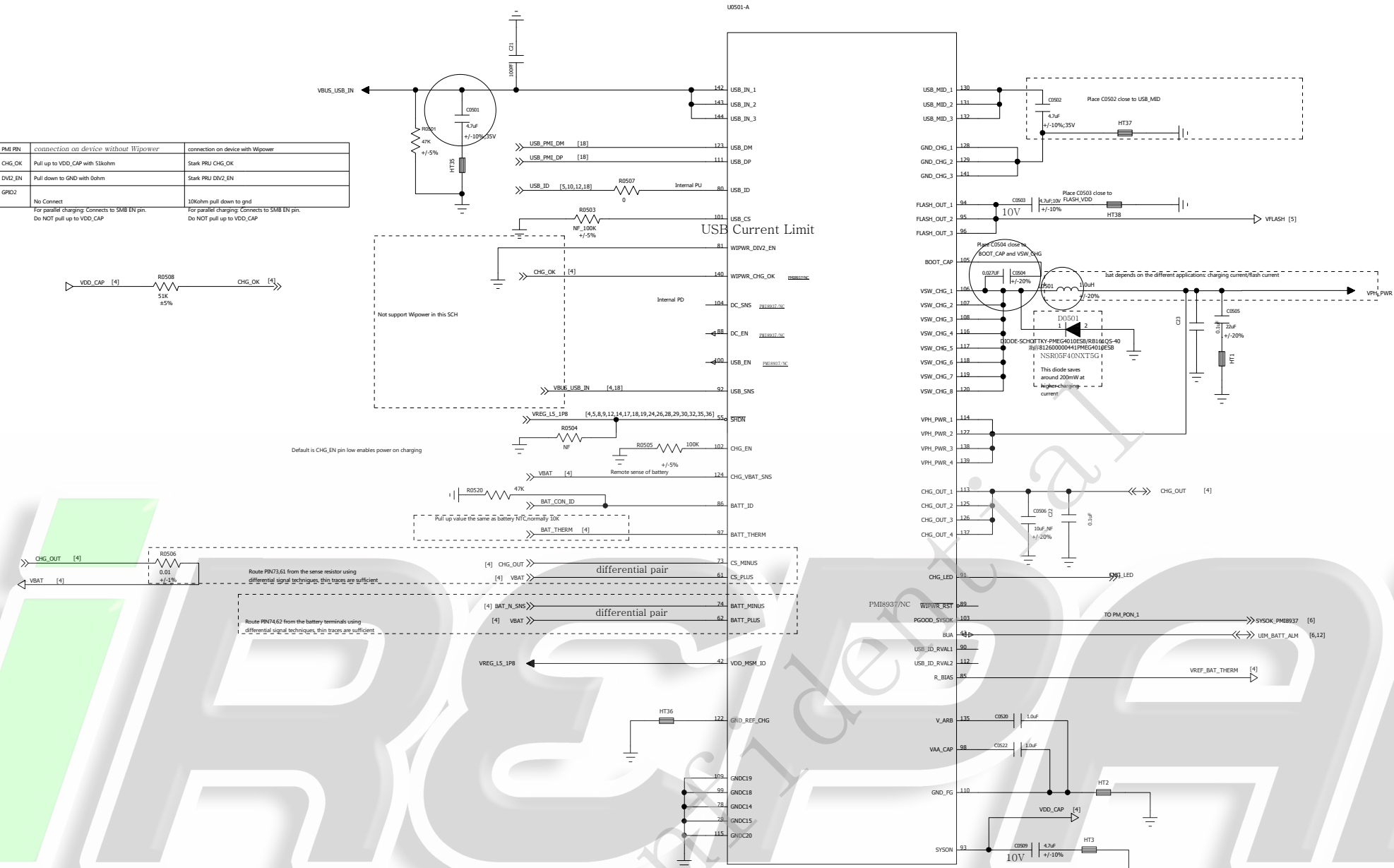
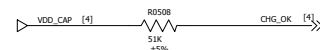
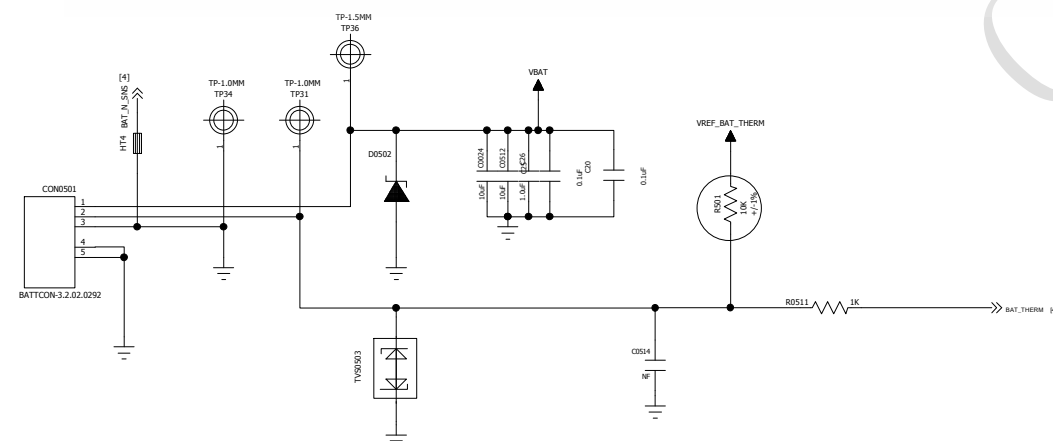


PMI PIN	connection on device without Wipower	connection on device with Wipower
CHG_OK	Pull up to VDD_CAP with 51kohm	Stark PRU CHG_OK
CHG_EN	Pull down to GND with 60hm	Stark PRU CHG_EN
GPIO2	No Connect For parallel charging Connects to SMB EN pin. Do NOT pull up to VDD_CAP	10Kohm pull down to gnd For parallel charging Connects to SMB EN pin. Do NOT pull up to VDD_CAP



PMI_VDD_CAP net is 5VDC, cap is 0603.
Do not use 0402 as may derate to -80%.

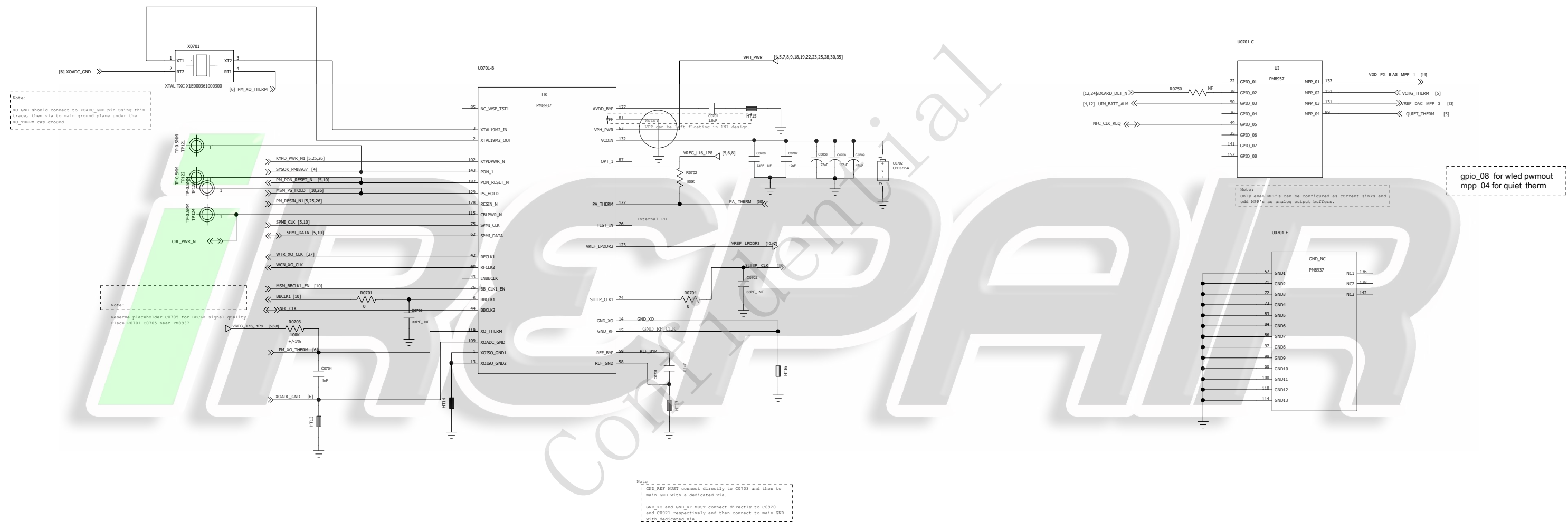


BATID Resistor Pull down to GND in BAT Module.
BAT_THERM NTC(68K B=4250) Pull down to GND in BAT Module.

BATTERY CONNECTOR

PMI8940 CHARGER

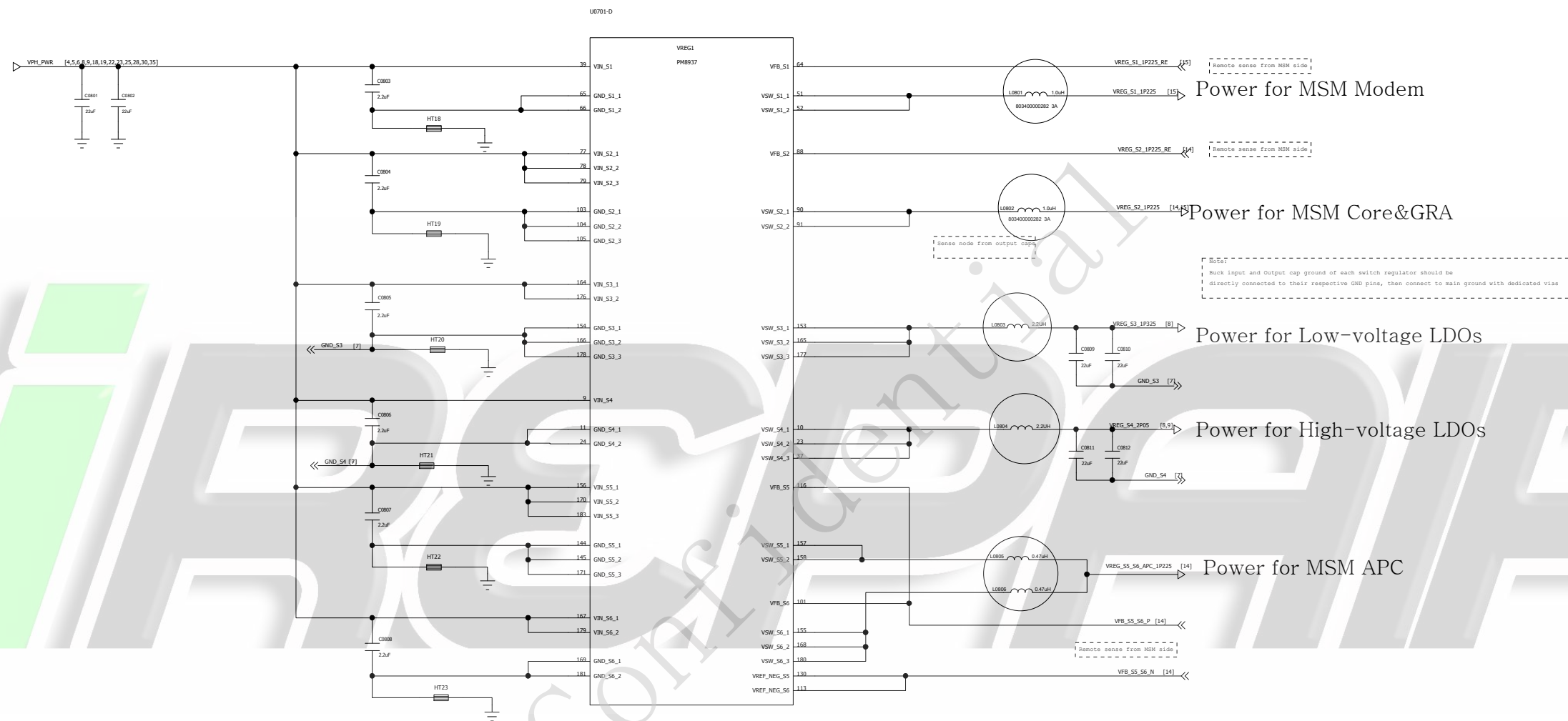
Title		
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Date:	Sheet	of



PM8937 Control and MPP/Clock

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Title
Sheet
Size Name Rev
Date: Sheet of

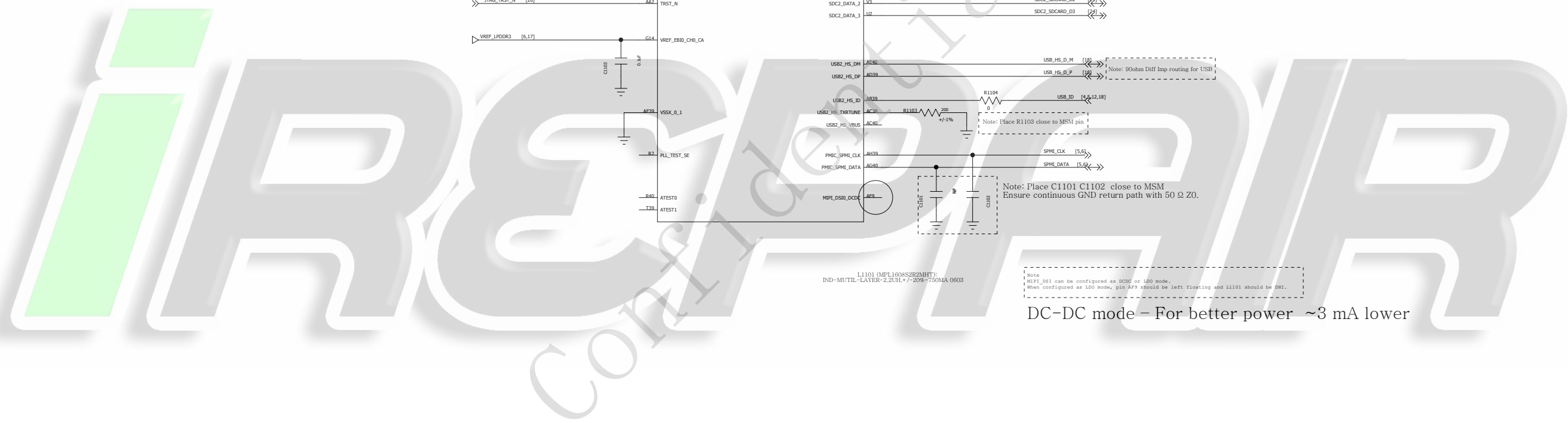


PM8937 Buck converter

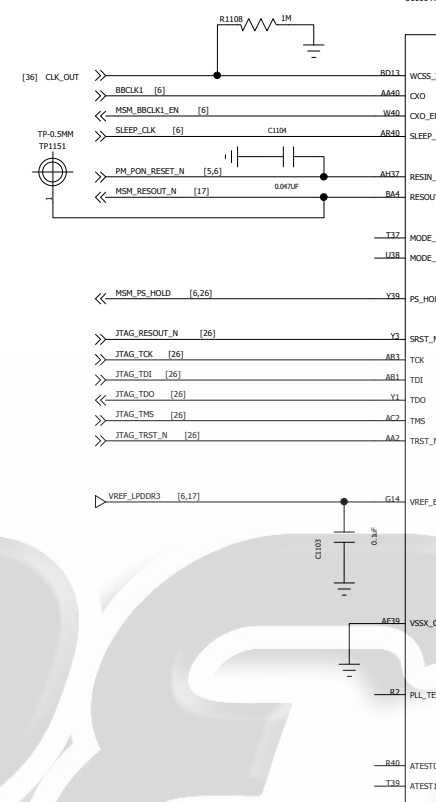
QUALCOMM Technologies, Inc.
12060 Torrey Pines Drive
San Diego, CA 92121-1714
U.S.A.

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Sheet	Rev
Size	Name
Date	Sheet of

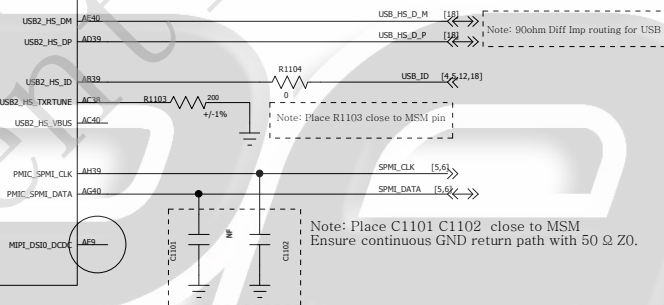
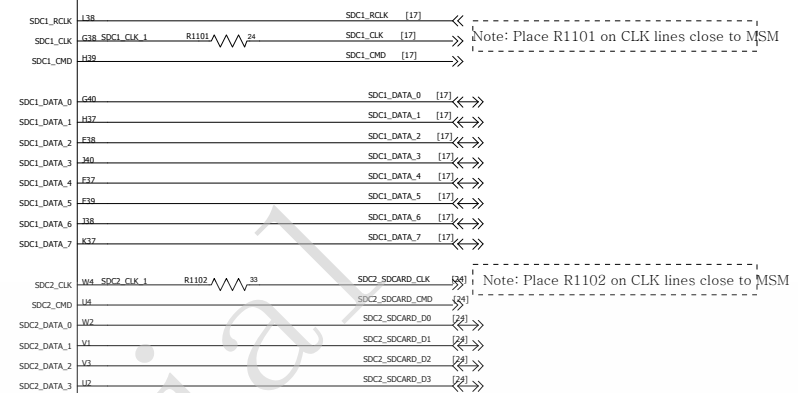




Note: WCSS_XO signal required only for 5GHZ.
When using WCN3615 connect pin BD13 to GND.



If eMMC 4.51 or earlier is used,
configure the unused MSM SDC1_RCLK pin to internal pull-down to prevent the bus from floating.

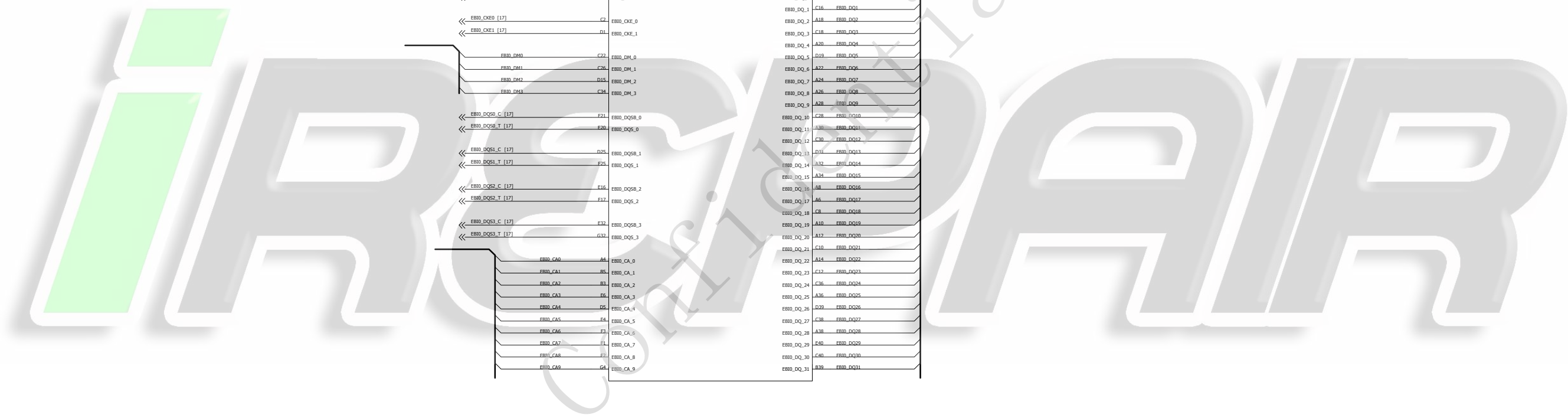


IND-MULTI-LAYER: 2.2UH +/-20% 750MA 0603
L1101 (MPT1608S282MHT)
Note: WCFI_001 can be configured as DCDC or LDO mode.
When configured as LDO mode, pin AP9 should be left floating and L1101 should be DMI.

DC-DC mode – For better power ~3 mA lower

MSM8917 Control

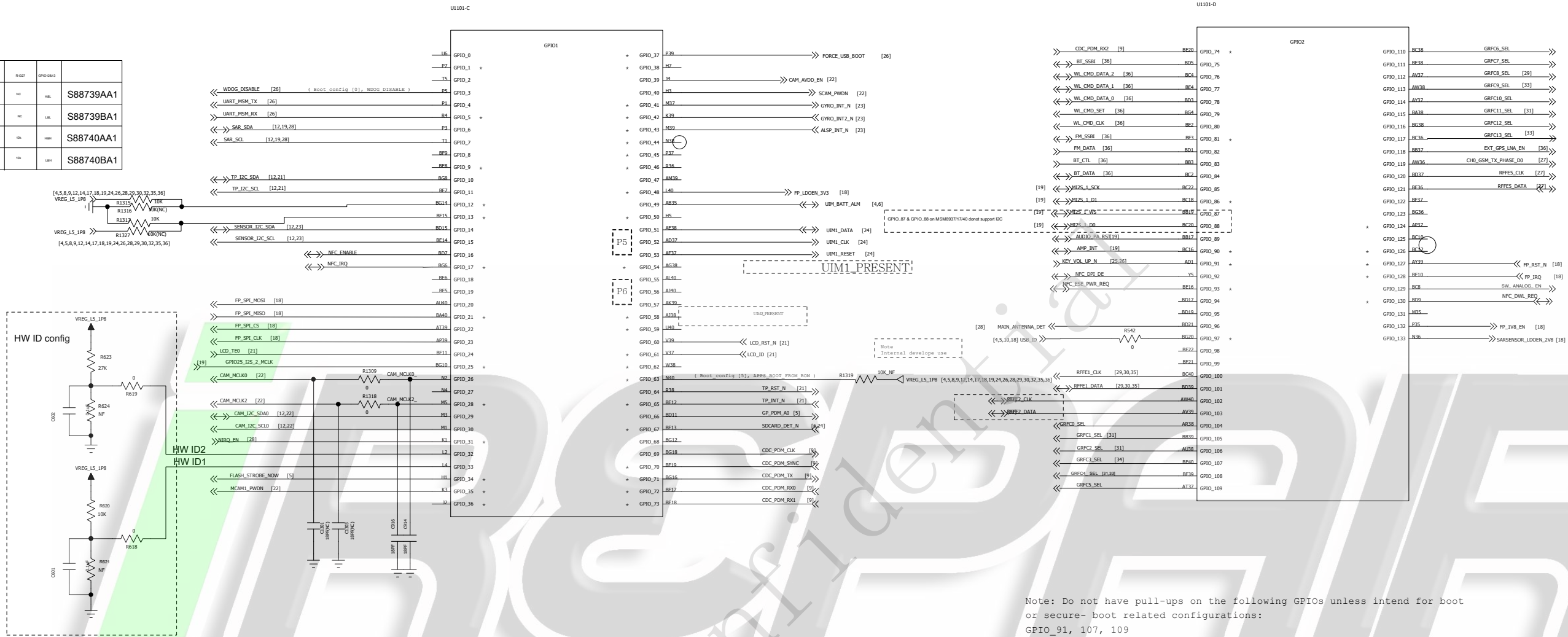
QUALCOMM Technologies, Inc. 3865 La Jolla Village Drive San Diego, CA 92121-1714 U.S.A.		
Title		
Sheet	Rev	
Size	Name	
Date	Sheet	of



MSM8917 EBI

Note: Asterisks (*) indicate modem power management (MPM) wake-up pins

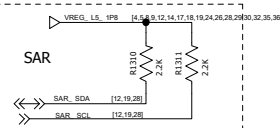
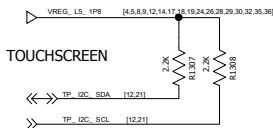
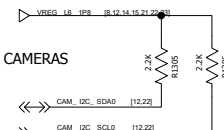
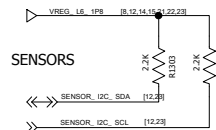
R1315	R1316	R1317	R1318	GPIO10813	
NA	NA	NA	NA	NA	S88739AA1
NA	NA	NA	NA	NA	S88739BA1
NA	NA	NA	NA	NA	S88740AA1
NA	NA	NA	NA	NA	S88740BA1



Note: Do not have pull-ups on the following GPIOs unless intend for boot or secure- boot related configurations:
GPIO_91, 107, 109

I2C PULL-UP RESISTORS

Note: Ensure SW sets these GPIOs (Sensor, CTP and Camera I2C bus) to inout pull down when the peripherals are powered off to eliminate leakage.

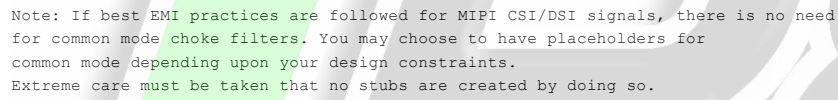


NFC

MSM8917 GPIO

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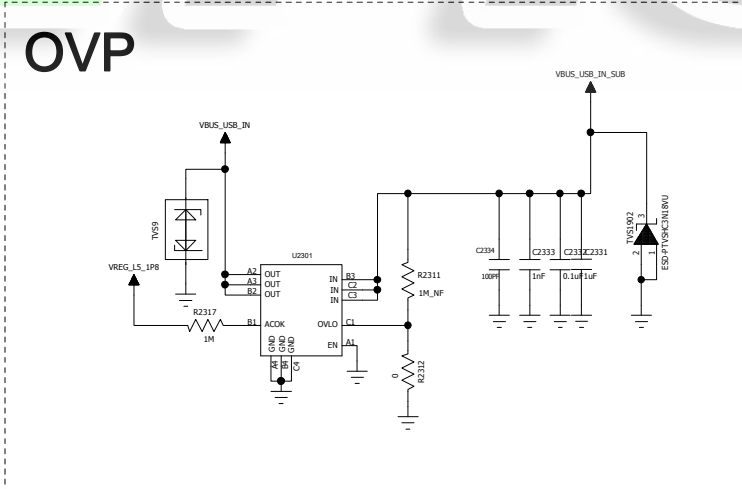
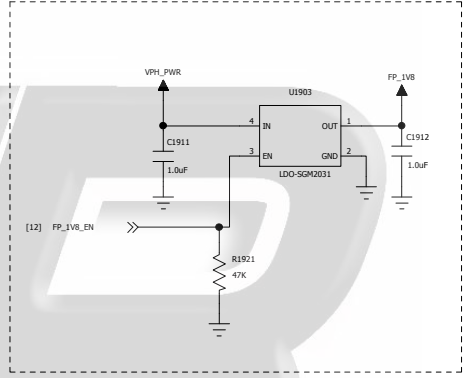
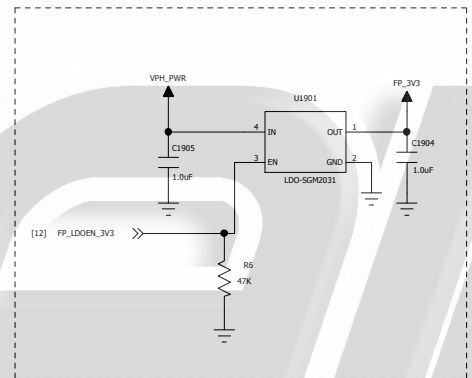
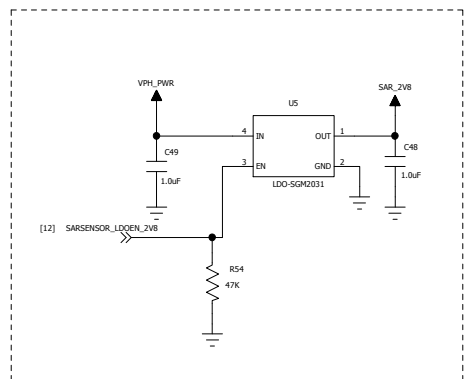
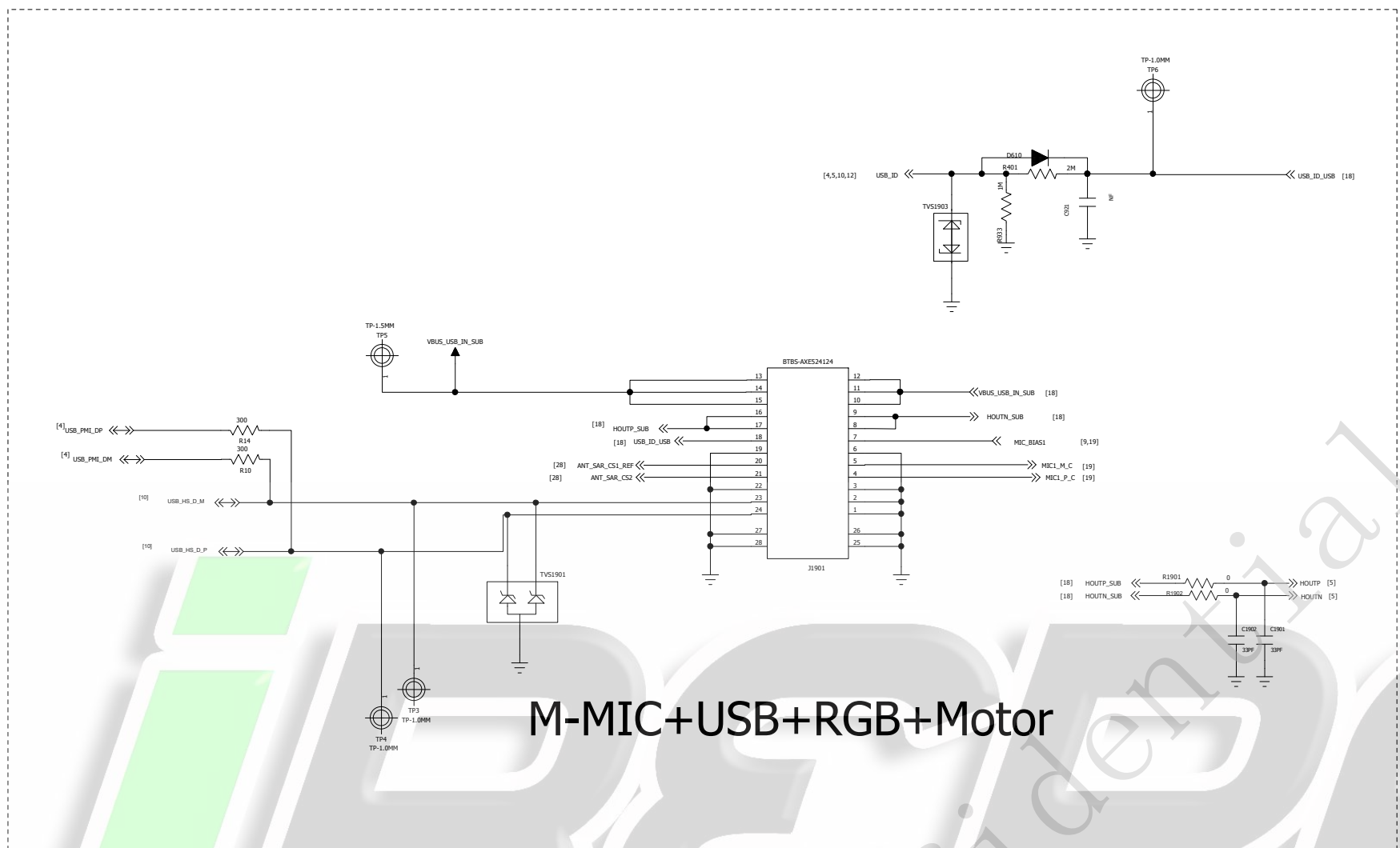
QUALCOMM Technologies, Inc.
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U.S.A.

Title Sheet	
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Date:	Sheet of

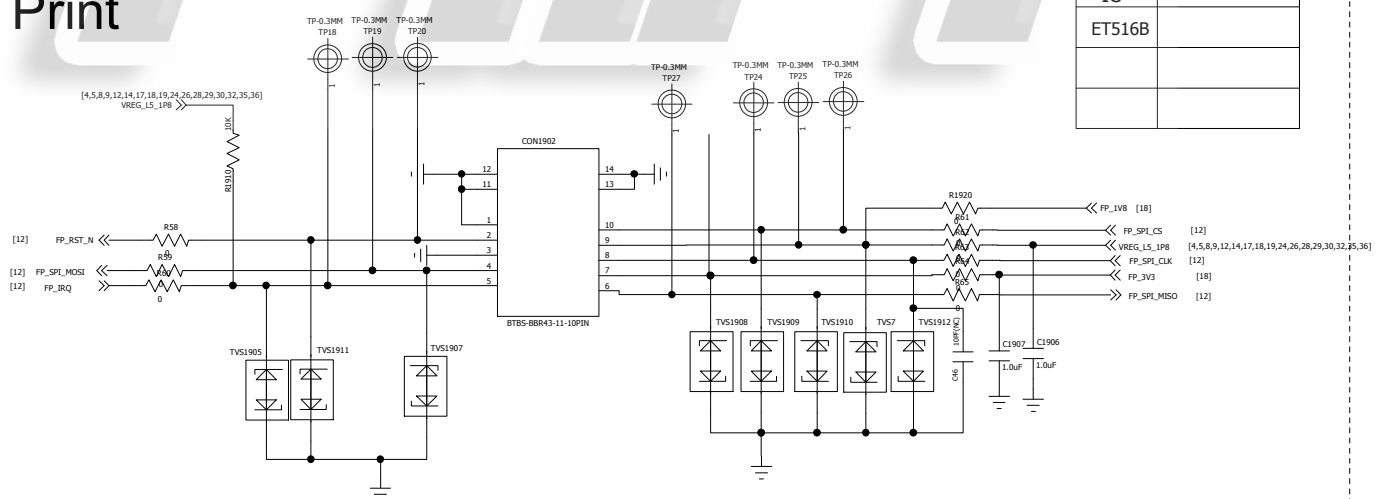


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Title		
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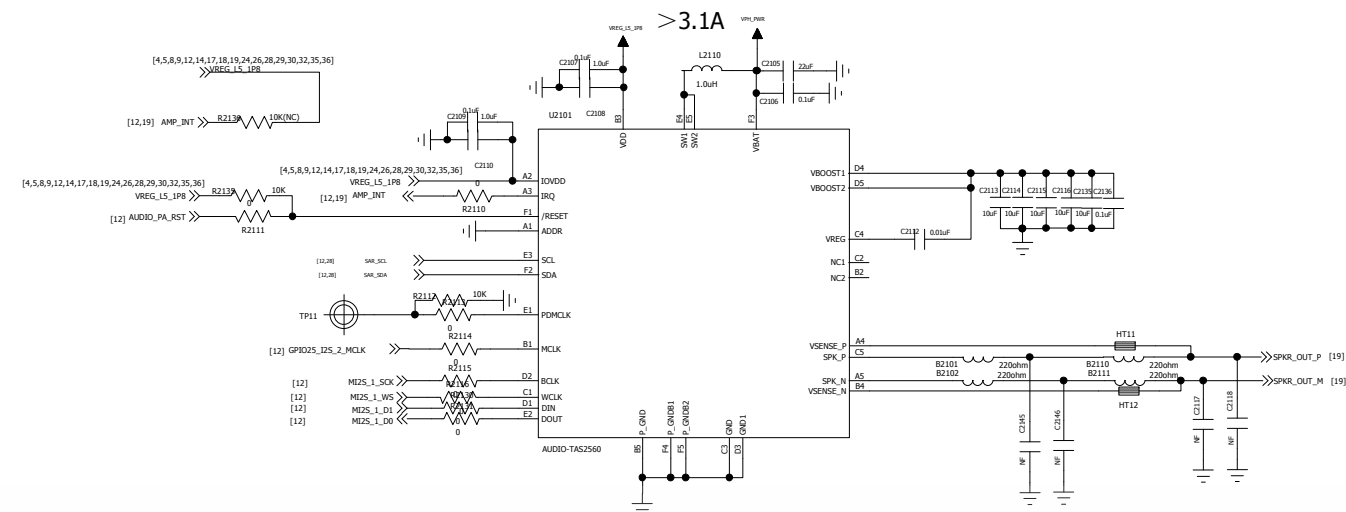


Finger Print

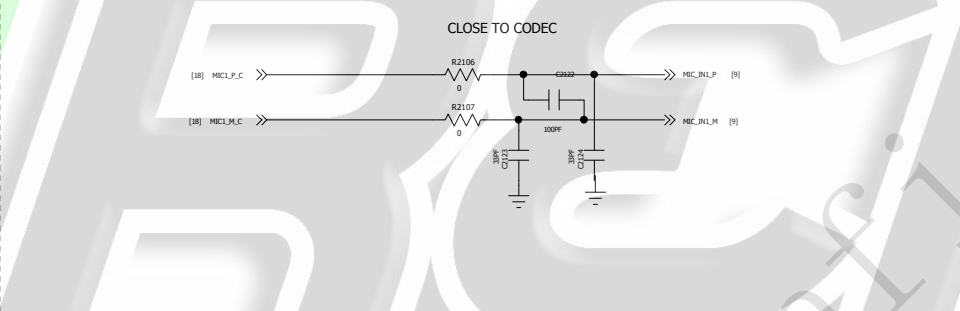


IC	
ET516B	

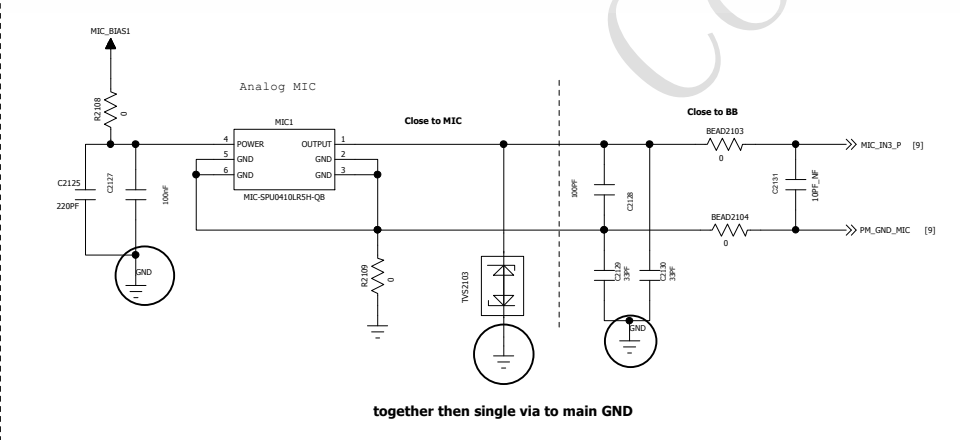
SMART PA



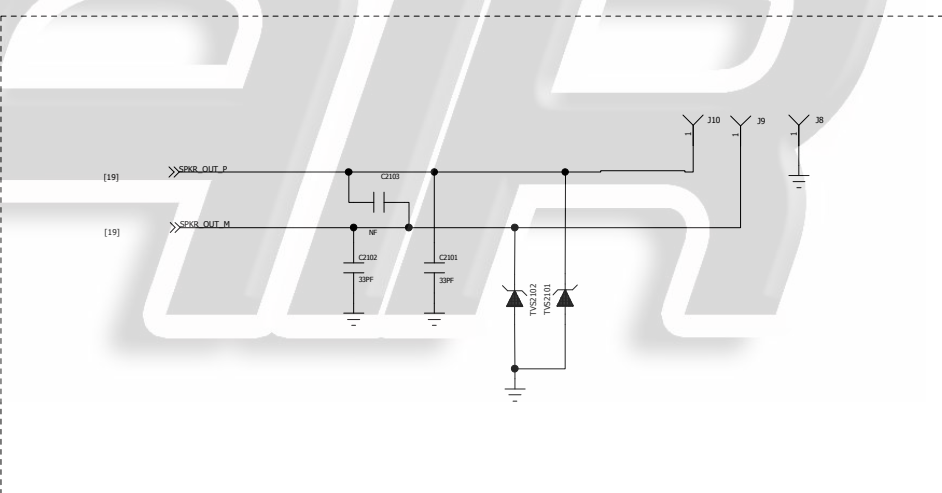
PRIMARY MIC



AUXMIC

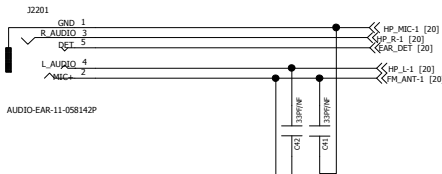


RECEIVER&speaker



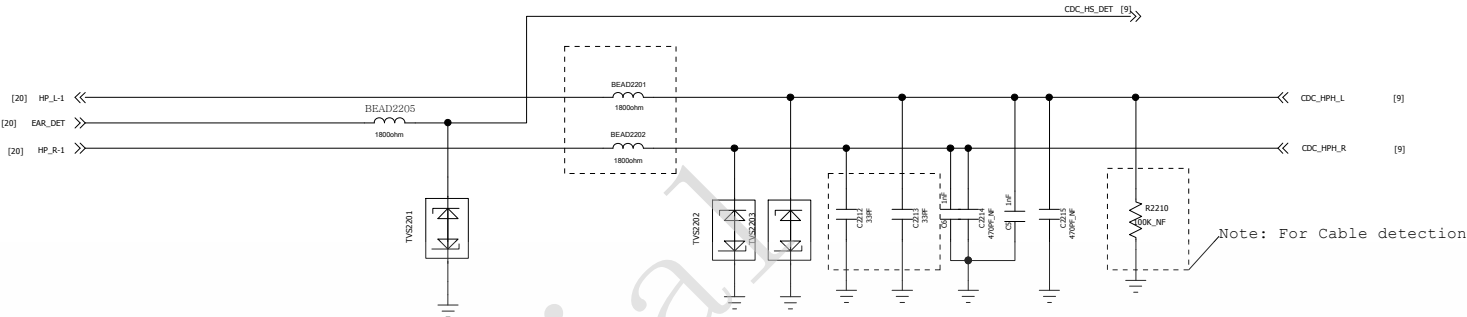
SECONDARY MIC/RECEIVER

AUDIO JACK

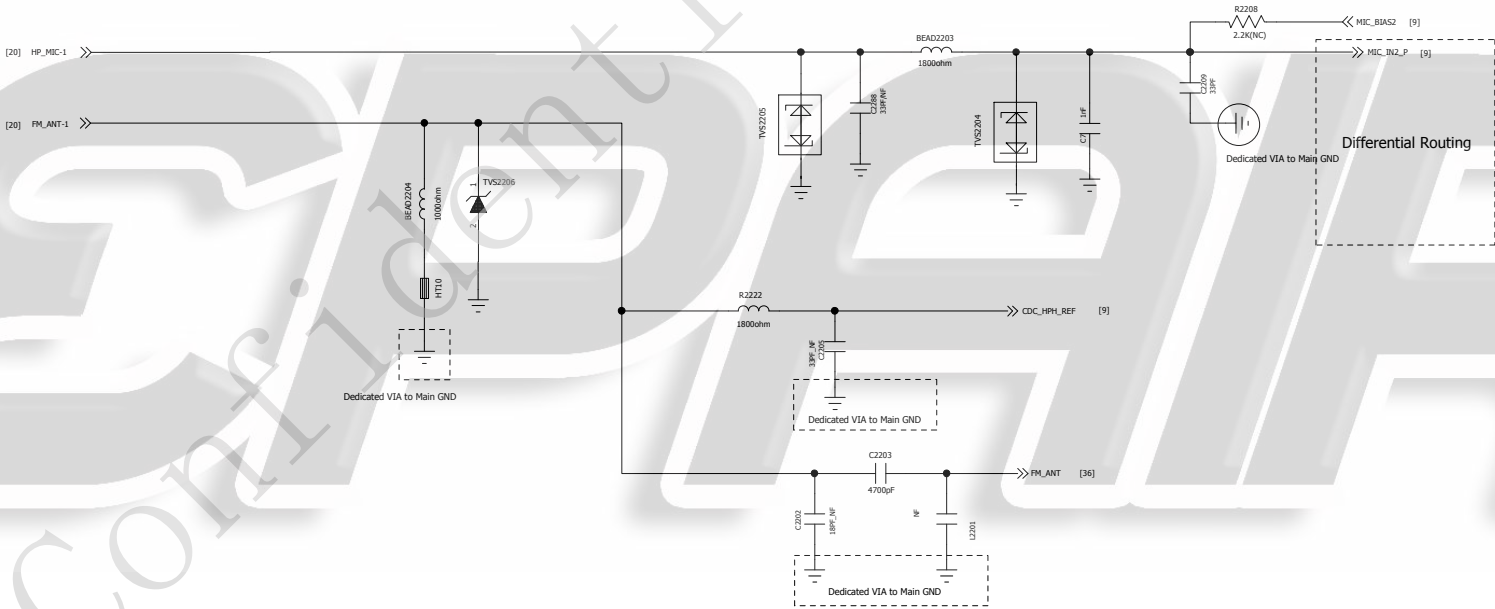


American Standard
HEADSET PLUG IN:HP_JACKDET "0"
HEADSET PLUG OUT:HP_JACKDET "1"

Note: Ferrite beads and their corresponding bypass capacitors on CDC_HPH_L, CDC_HPH_R and CDC_HPH_REF are needed to reduce noise generated by audio/FM concurrency



Note: For Cable detection



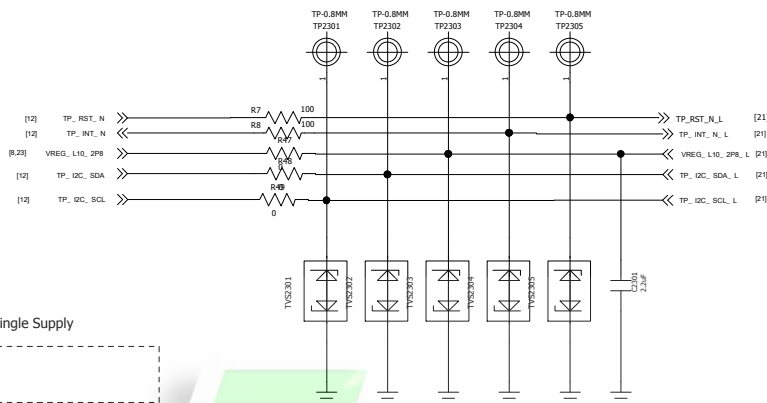
Differential Routing

EARPHONE

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CTP



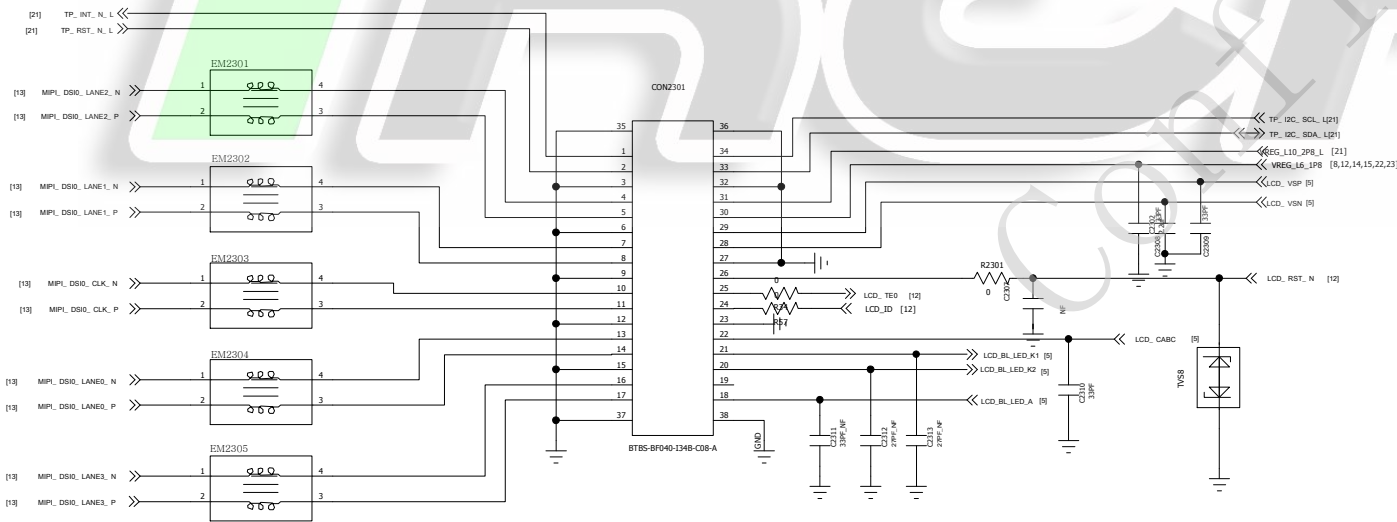
Power on sequence: Single Supply

FocalTech: FT5336

I2C Address

Write 0x70 It can be changed by FACTORY.
Read 0x71

LCD



dijin lcn id=1

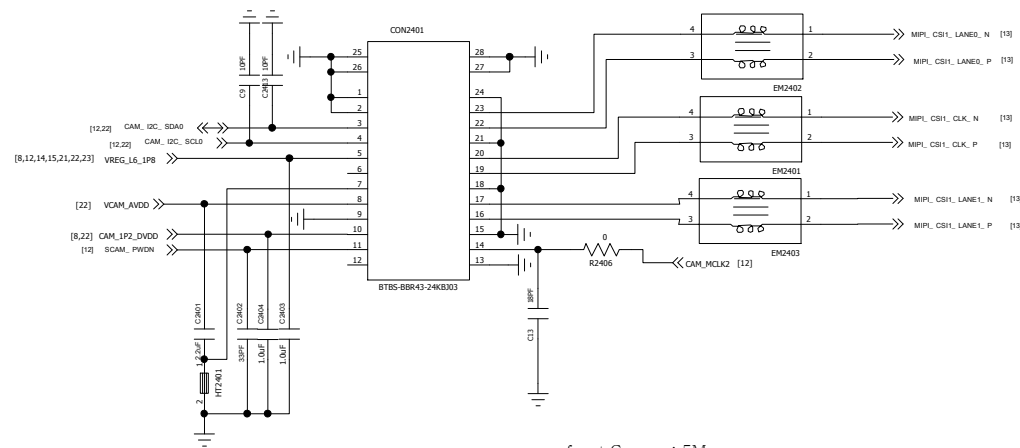
Note: If best EMI practices are followed for MIPI CSI/DSI signals, there is no need for common mode choke filters. You may choose to have placeholders for common mode depending upon your design constraints. Extreme care must be taken that no stubs are created by doing so.

LCD interface and backlight

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FRONT_camera



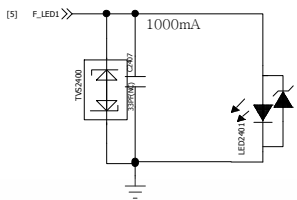
front Camera: 5M
Sensor: OV05675-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V

I2C Address	
Write	0x6C
Read	0x6D

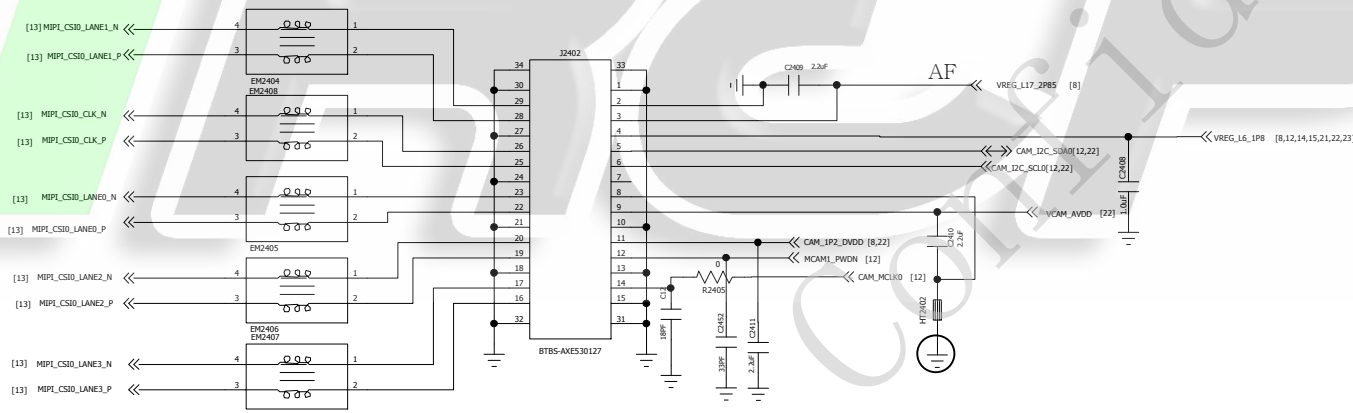
front Camera: 5M
Sensor: OV05675-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V

I2C Address	
Write	0x6C
Read	0x6D

FLASH LED(front)



Main Camera



Rear Camera: 8M
Sensor: OV08856-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18

I2C Address	
Write	0x20
Read	0x21

Rear Camera: 8M
Sensor: OV08856-GA4A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18

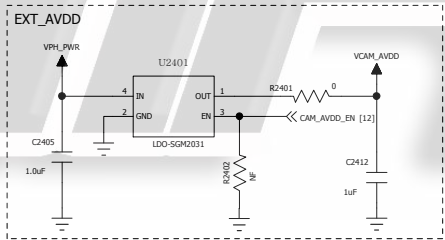
I2C Address	
Write	0x20
Read	0x21

Rear Camera: 13M
Sensor: OV13855-GA5A
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18

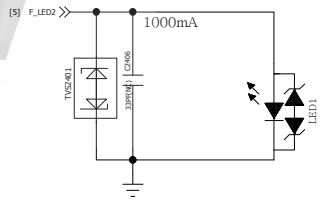
I2C Address	
Write	0x6C
Read	0x6D

Rear Camera: 13M
Sensor: S5K3L8XXM3-FGX9
AVDD:2.8V
DOVDD:1.8V
DVDD:1.2V
AFVDD:2.8V
Driver ID :0X18

I2C Address	
Write	0x5A
Read	0x5B



FLASH LED(back)

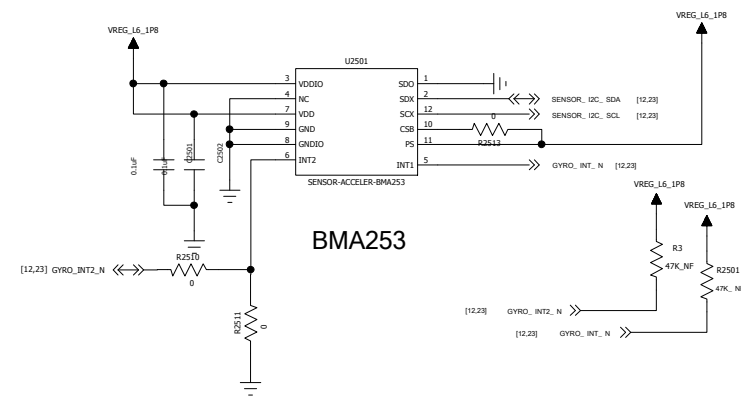


Main/Slave Camera and Flash

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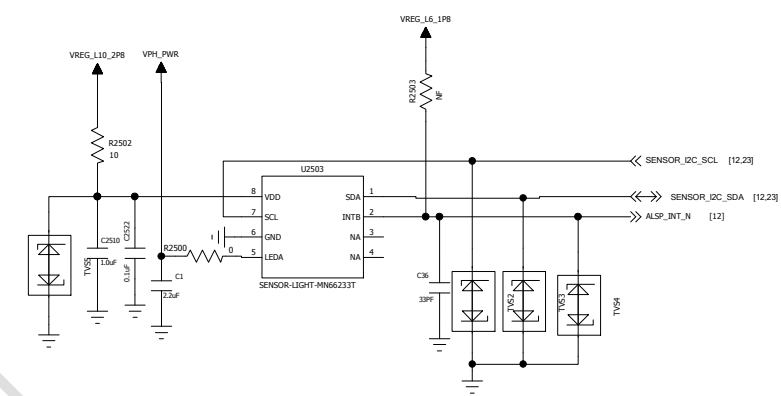
Title	
Sheet	Rev
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G-Sensor



	BM253	KXJT3_1058
R2510	OR	NC
R2511	NC	OR

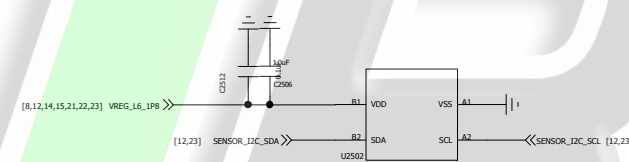
ALS+PS+IR



EMINENT:	
	I2C Address
Write	
Read	

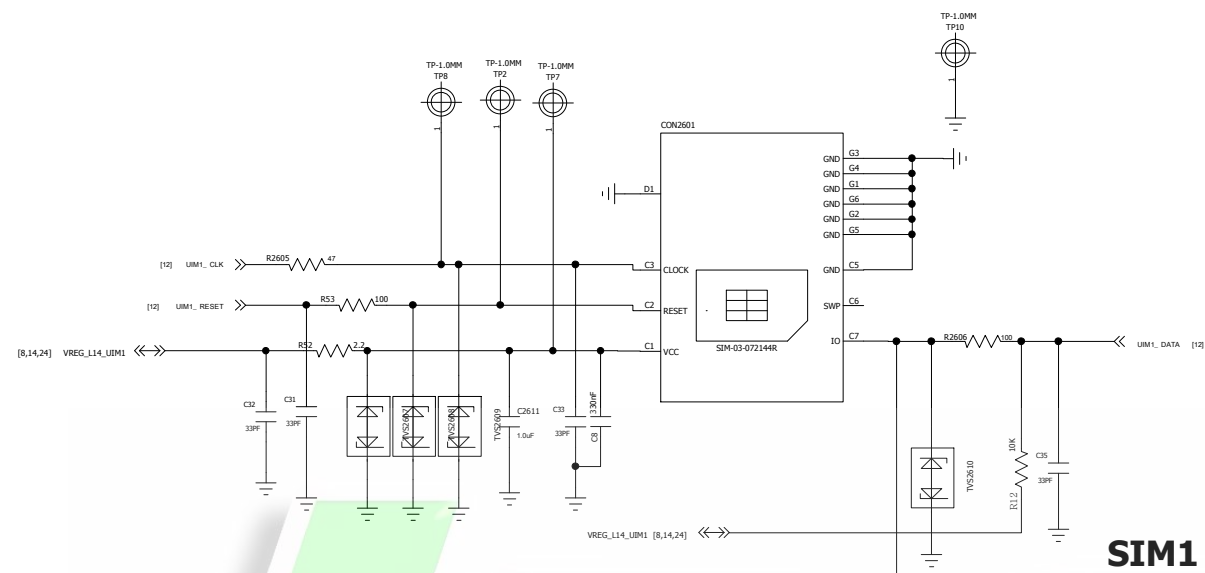
desense apk enable sensor when we enable the audio

M-Sensor

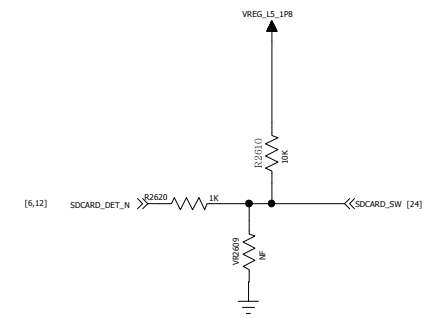


I2C Address	
Write	0x18
Read	0x19

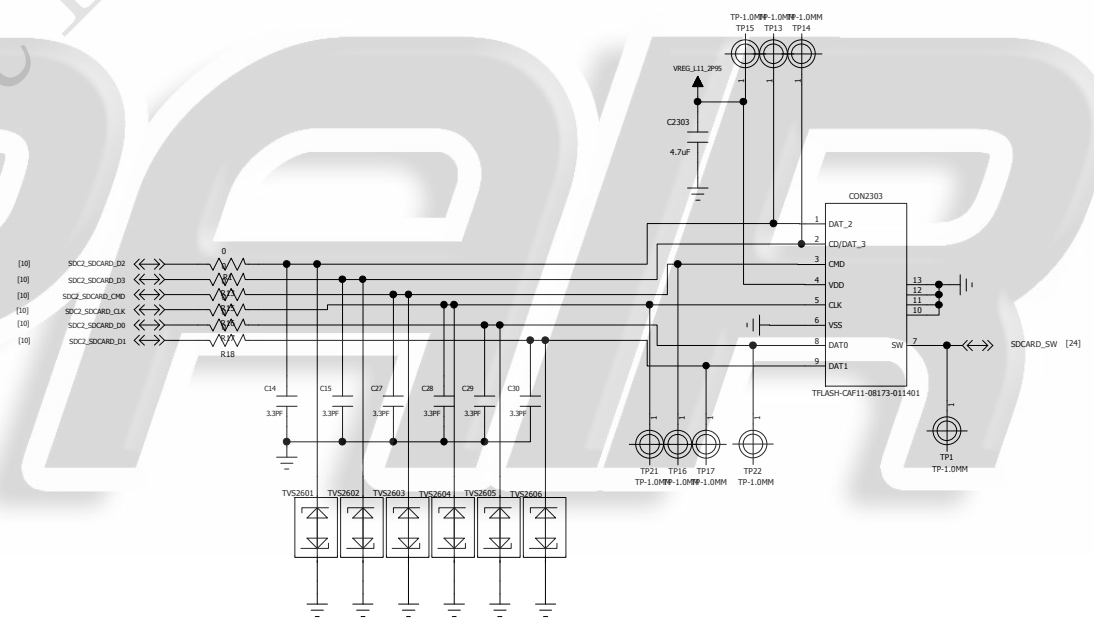
SENSORS



SIM1



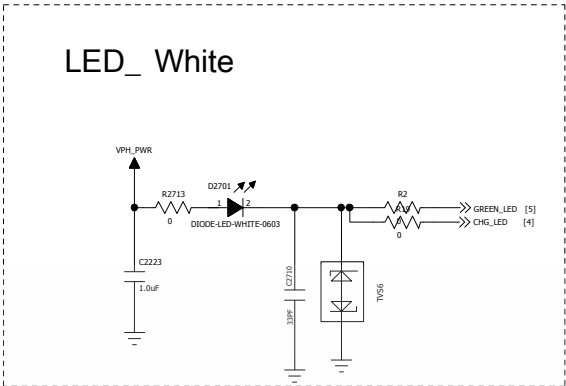
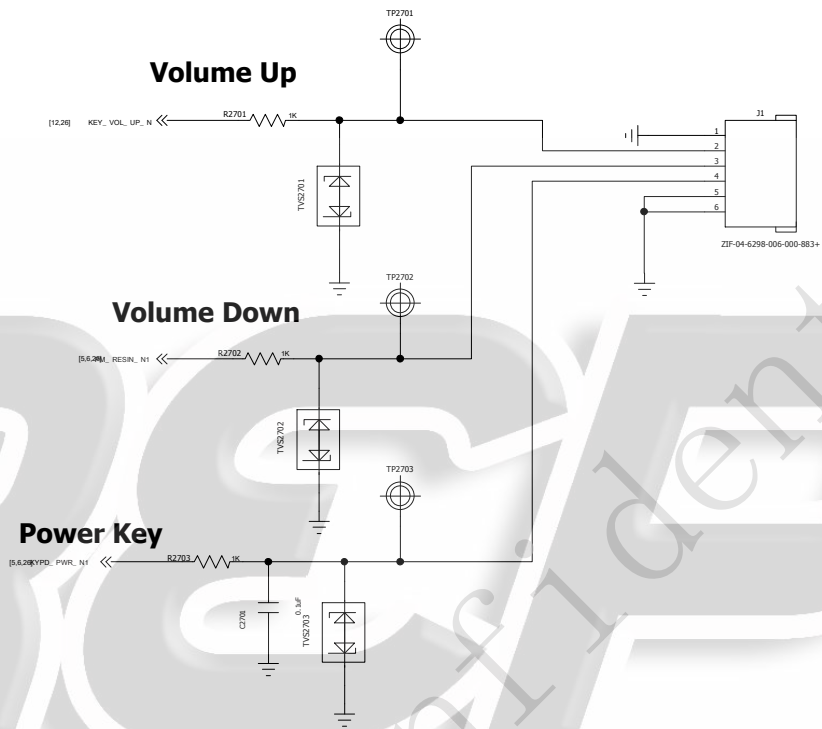
This int should give to GPIO54/GPIO58/GPIO67



TF

SIM/TF card

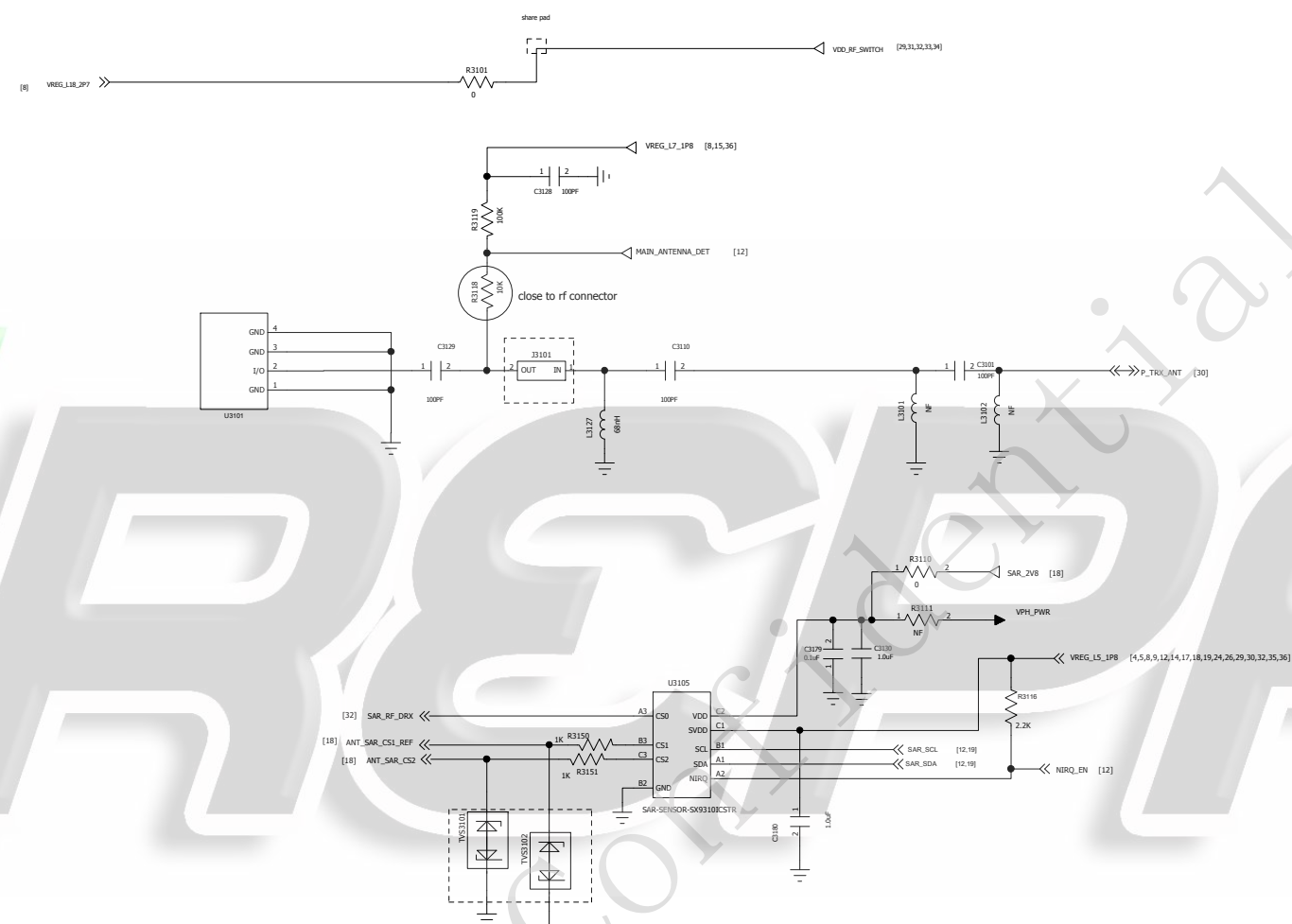
Signal	Description
KEY_VOL_UP_N	Volume Up
PM_RESIN_N	Volume Down
KYPD_PWR_N	POWER ON



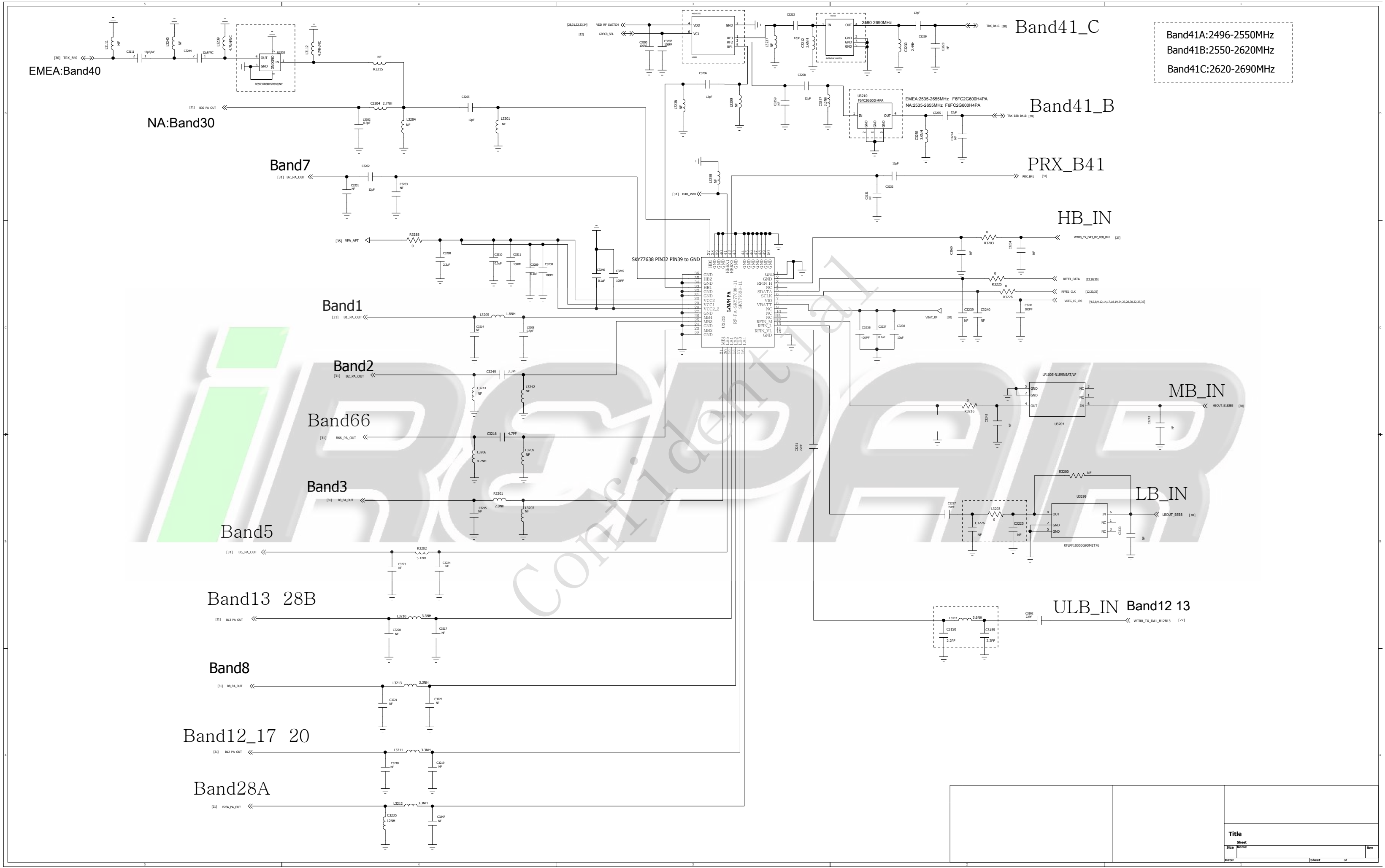
Keypad/LED/Status indicator

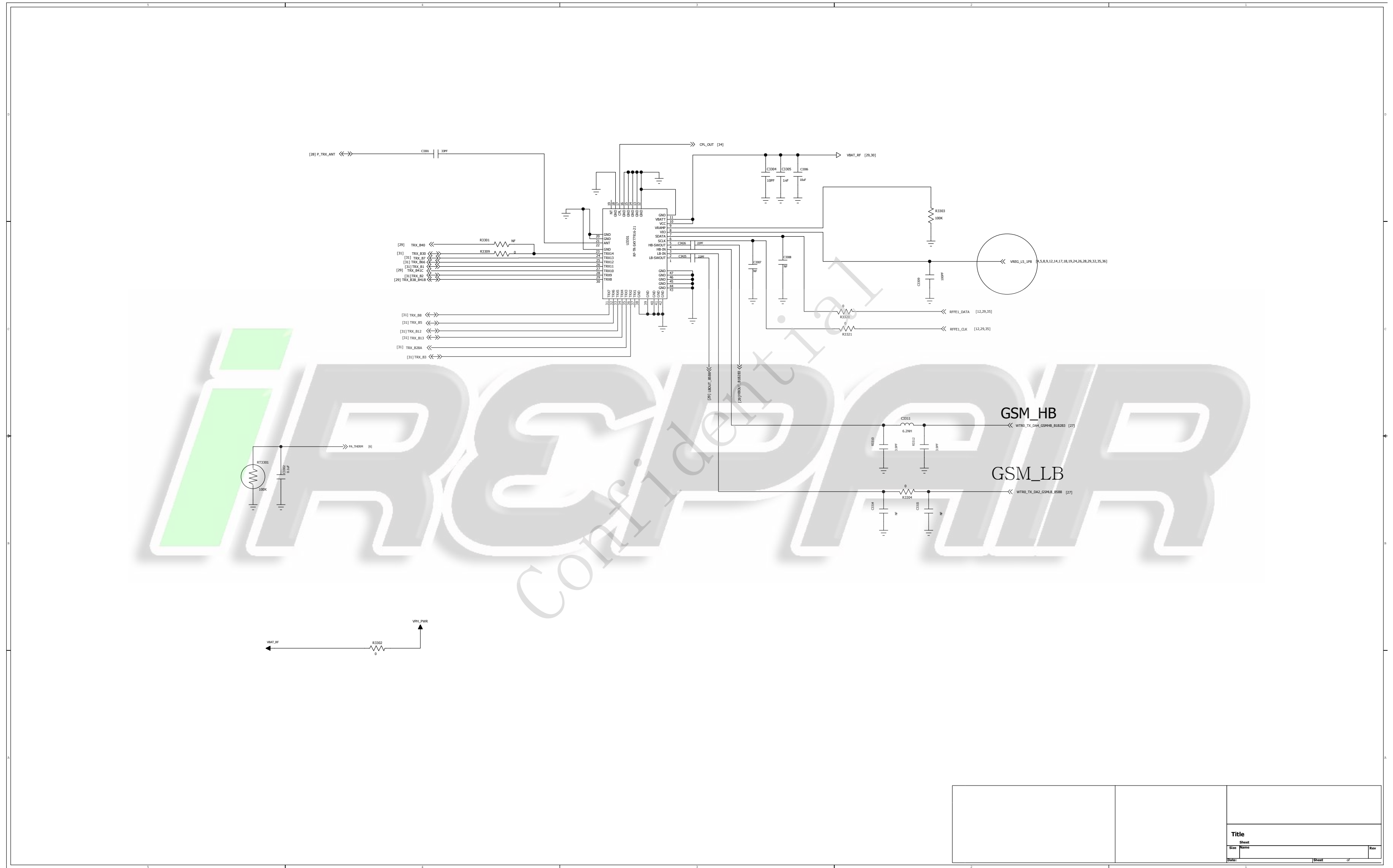
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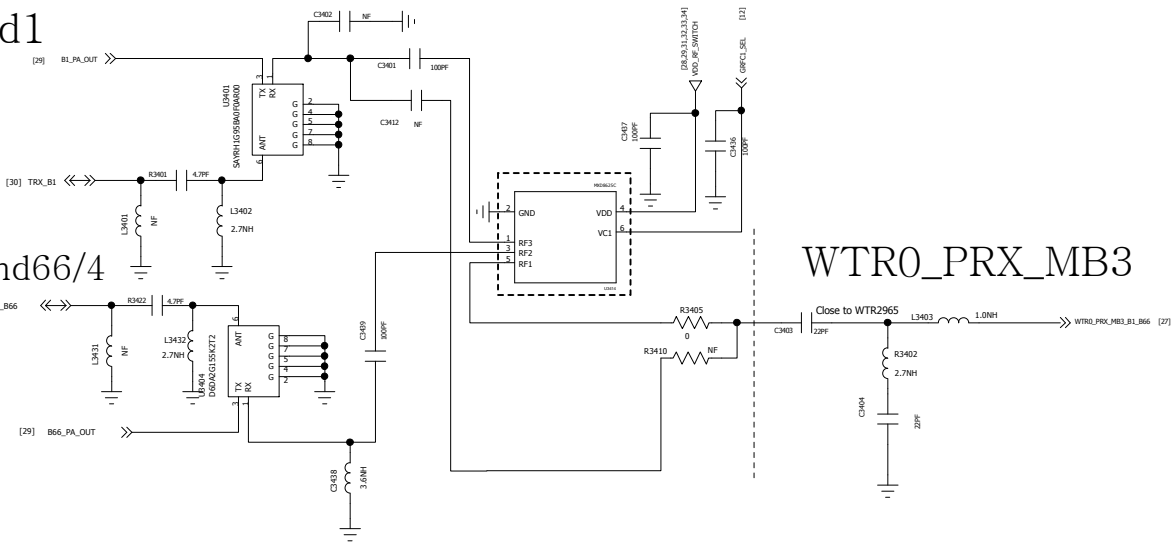
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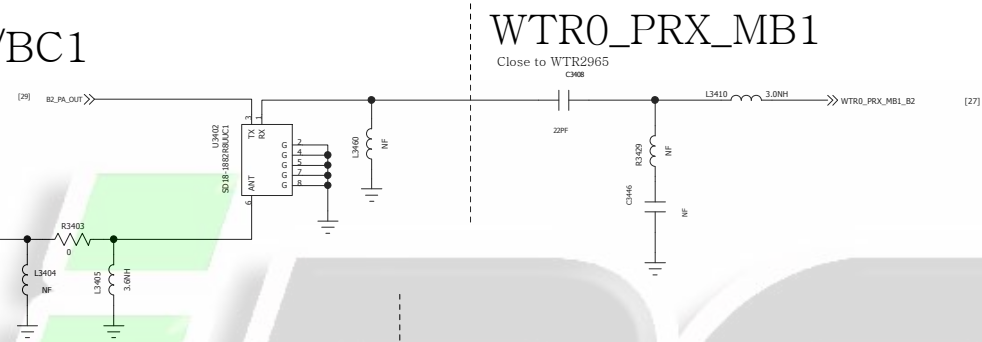
Band1

NA:Band66/4



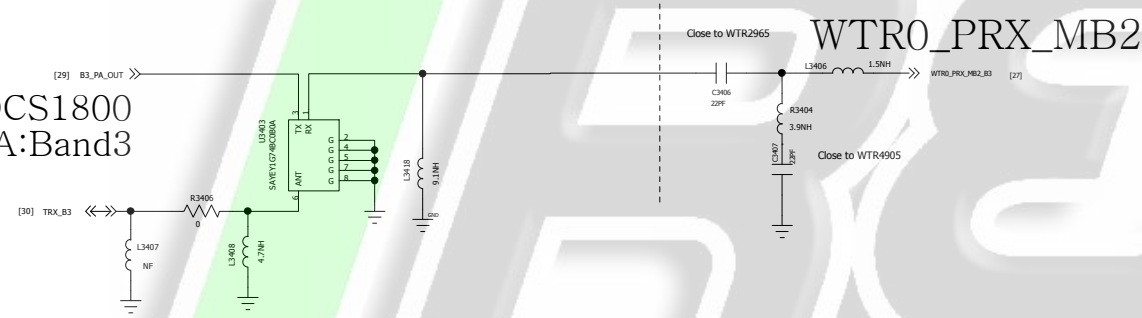
WTR0_PRX_MB3

NA:Band2/25/BC1
EMEA:WB2



WTR0_PRX_MB1

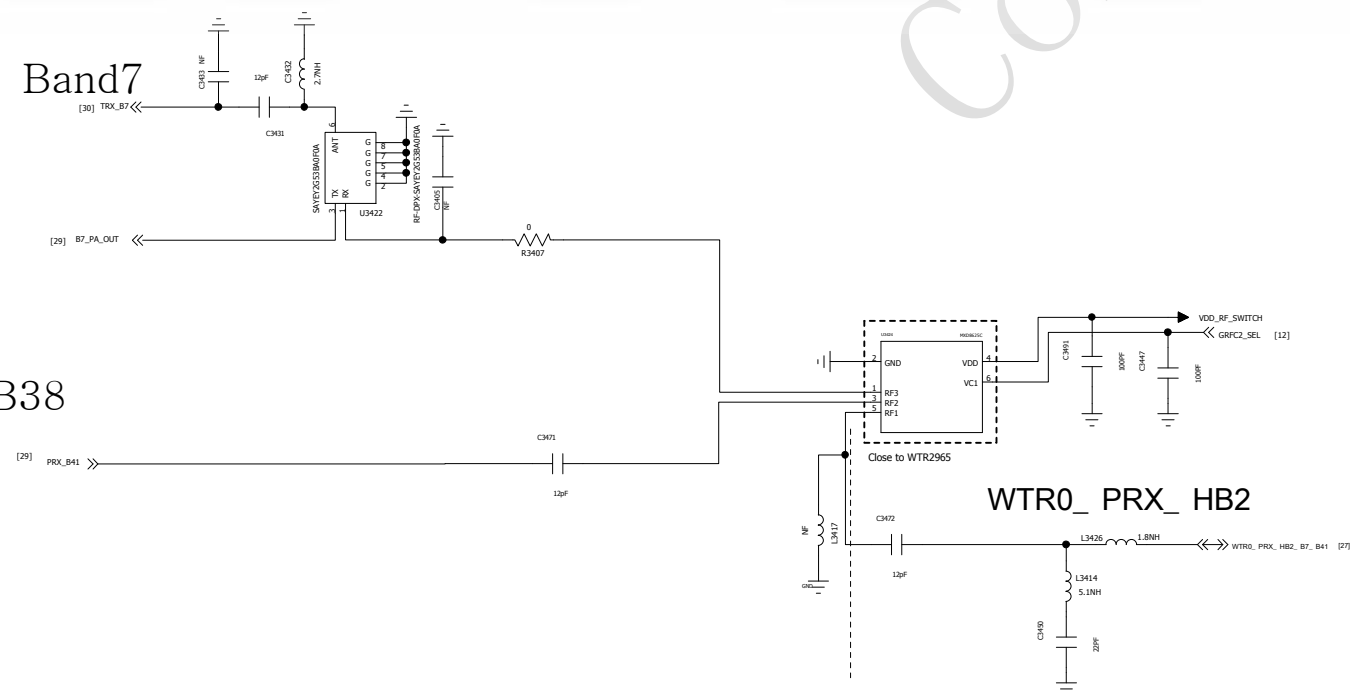
NA:DCS1800
EMEA:Band3



WTR0_PRX_MB2

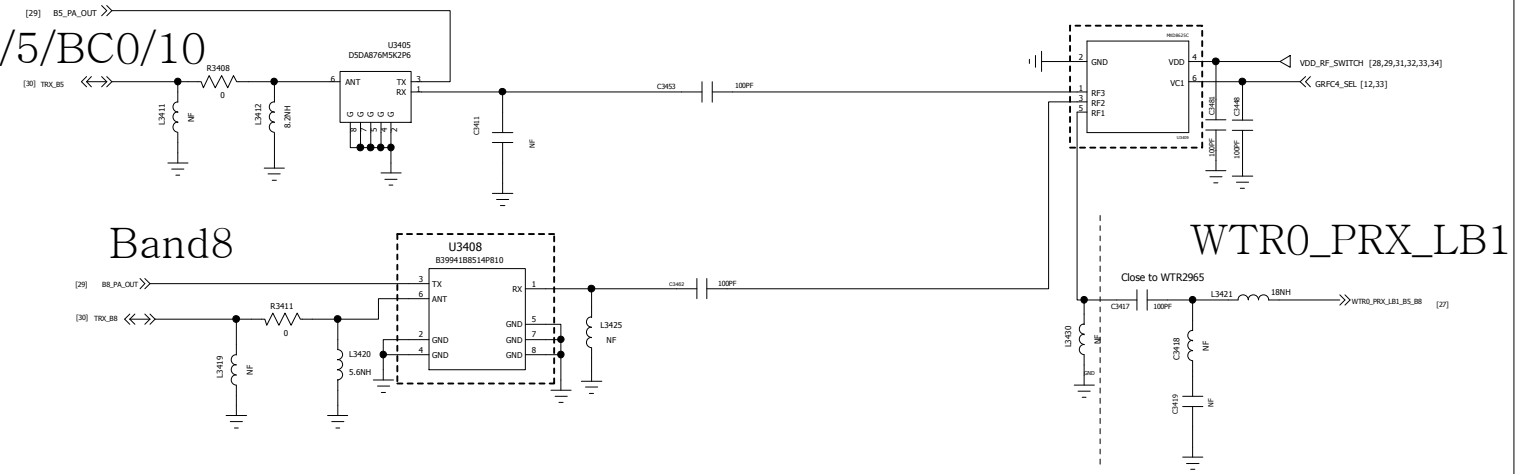
Band7

Band41_Full/B38



WTR0_PRX_HB2

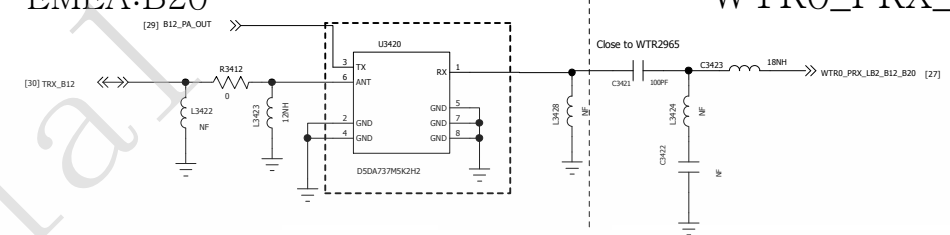
Band26/5/BC0/10



WTR0_PRX_LB1

Band8

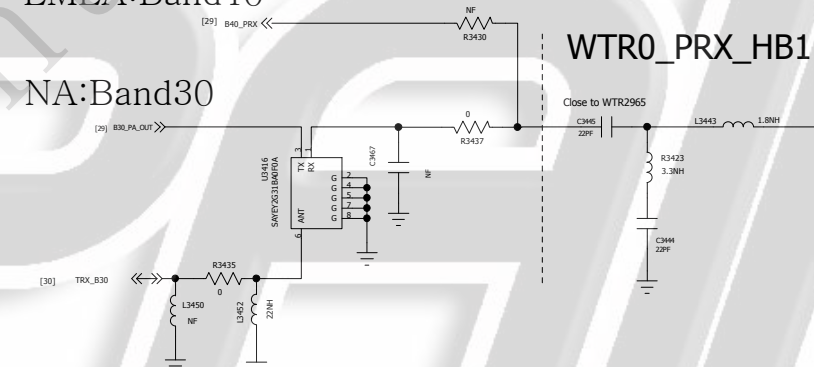
NA:Band12/17
EMEA:B20



WTR0_PRX_LB2

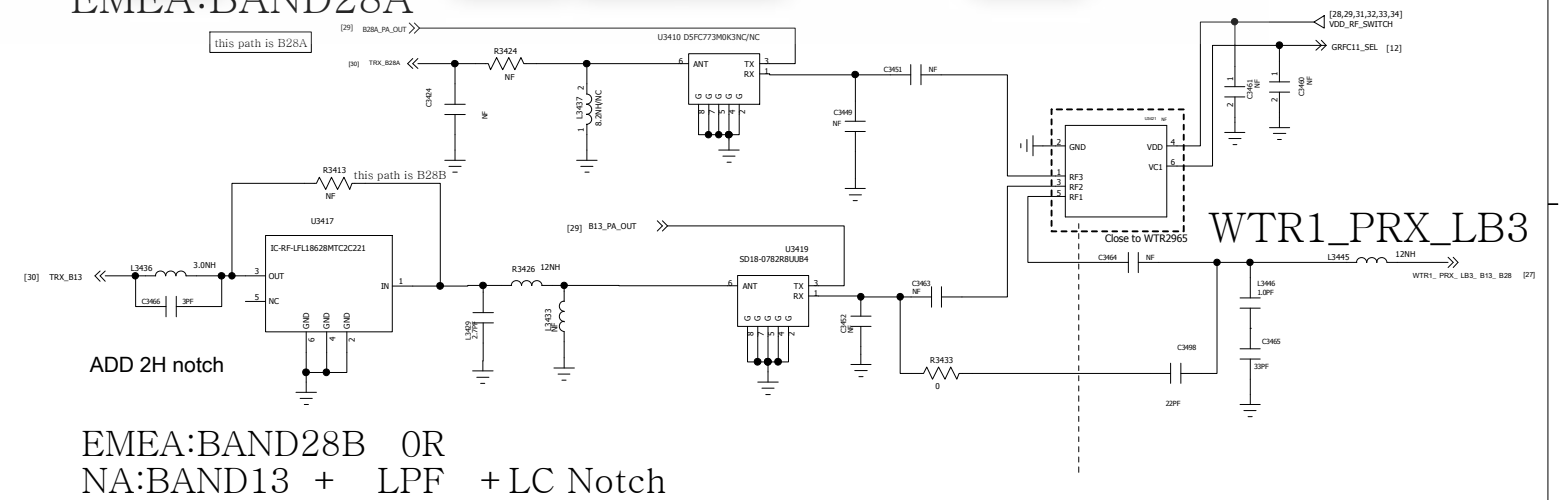
EMEA:Band40

NA:Band30



WTR0_PRX_HB1

EMEA:BAND28A



WTR1_PRX_LB3

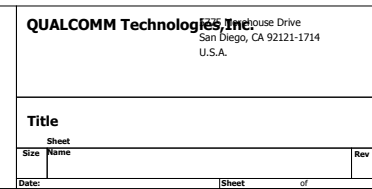
ADD 2H notch

EMEA:BAND28B OR
NA:BAND13 + LPF + LC Notch

Note:The matching need
close to WTR, RX ports have
DC at the pin, so it need DC block,

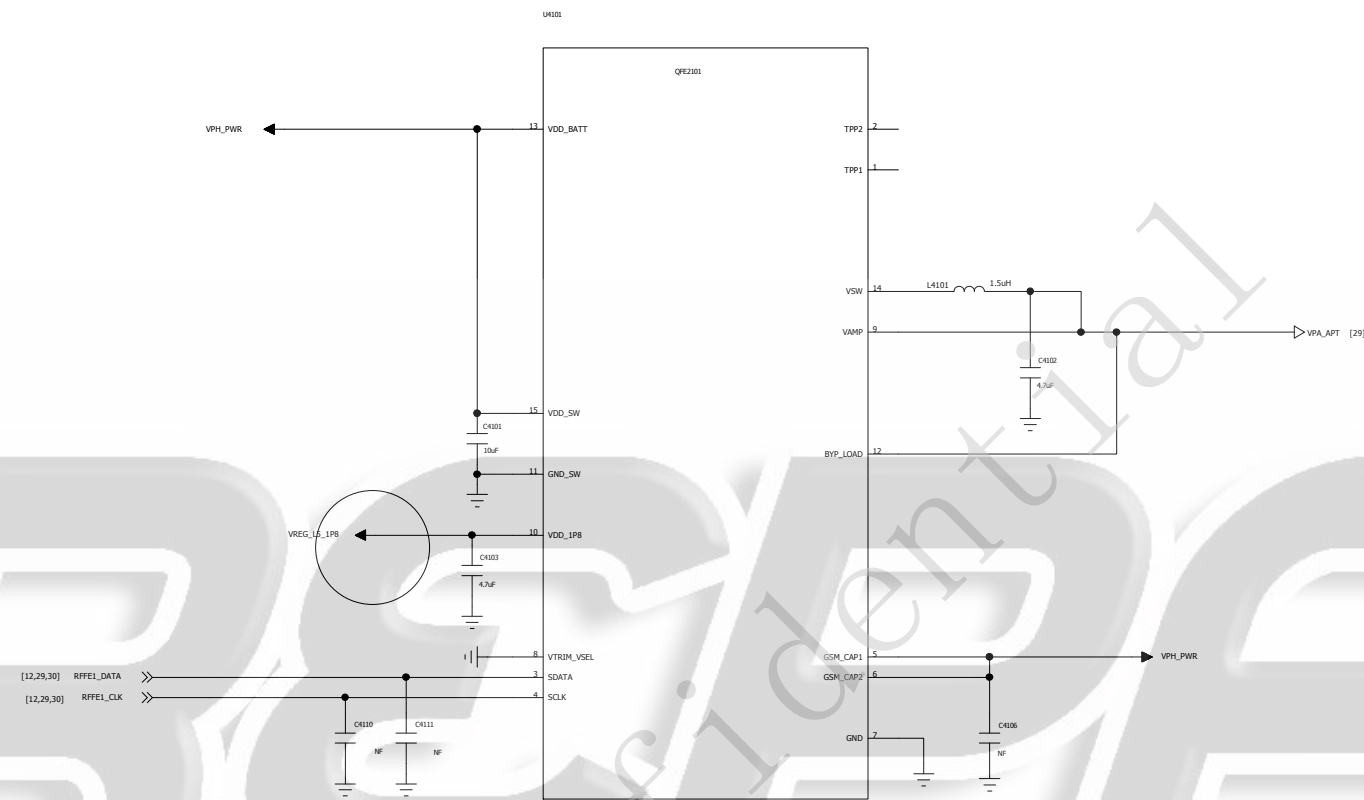


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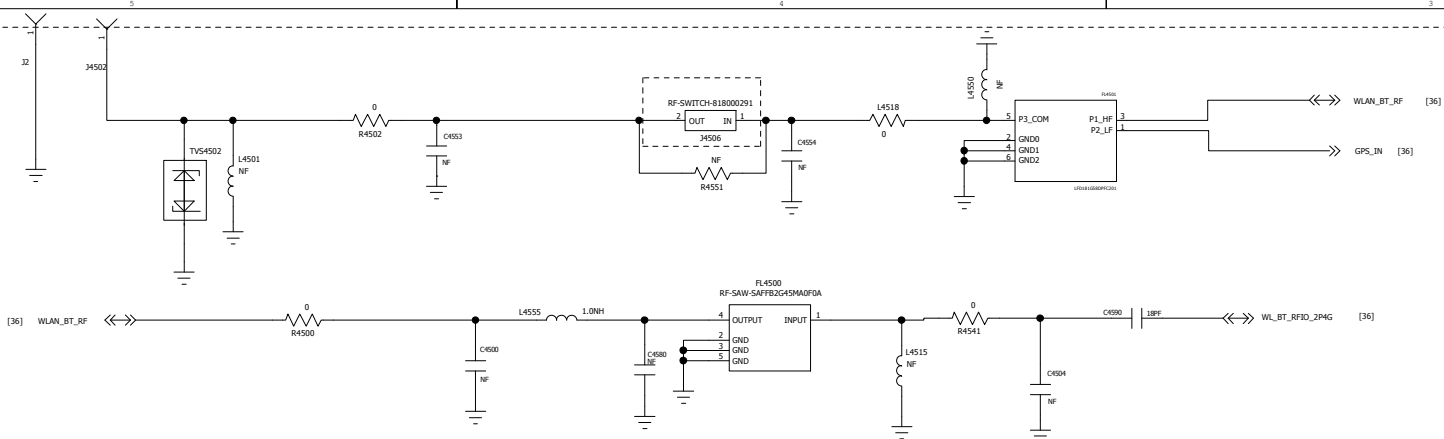


Note: Use a single low impedance power plane/fill VPA for all ET PA VCC1/VCC2.

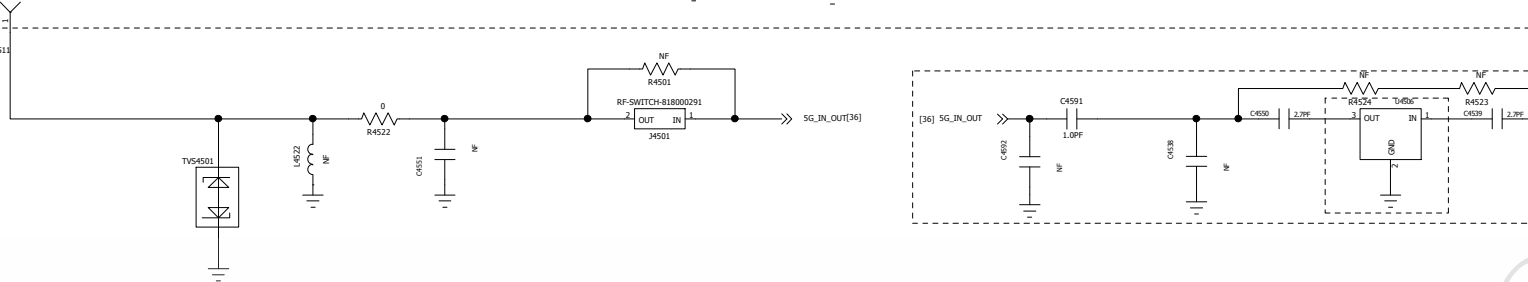
Refer to 80-NA681-91 rev.B or later revisions for more layout details.



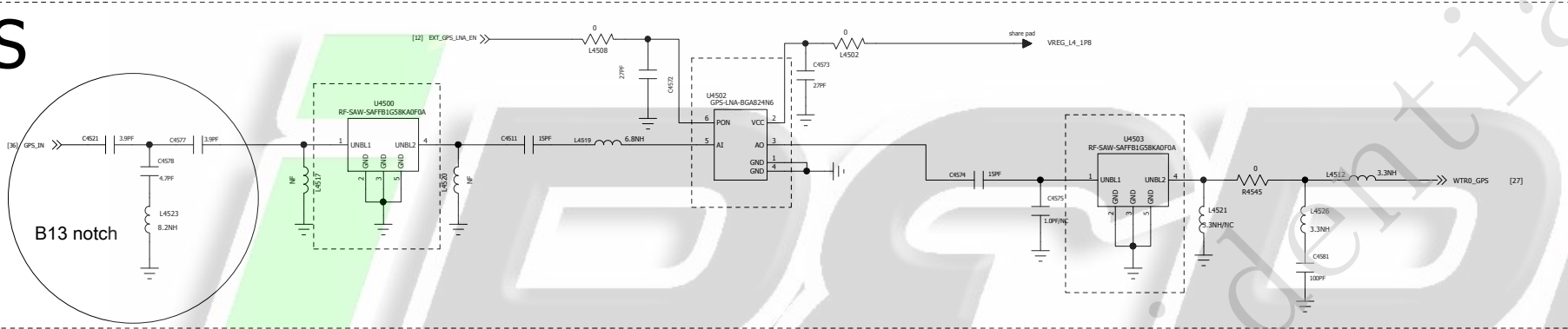
2.4G



5G

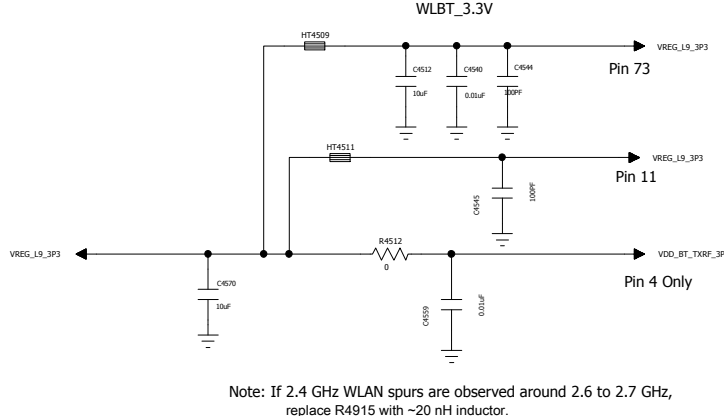
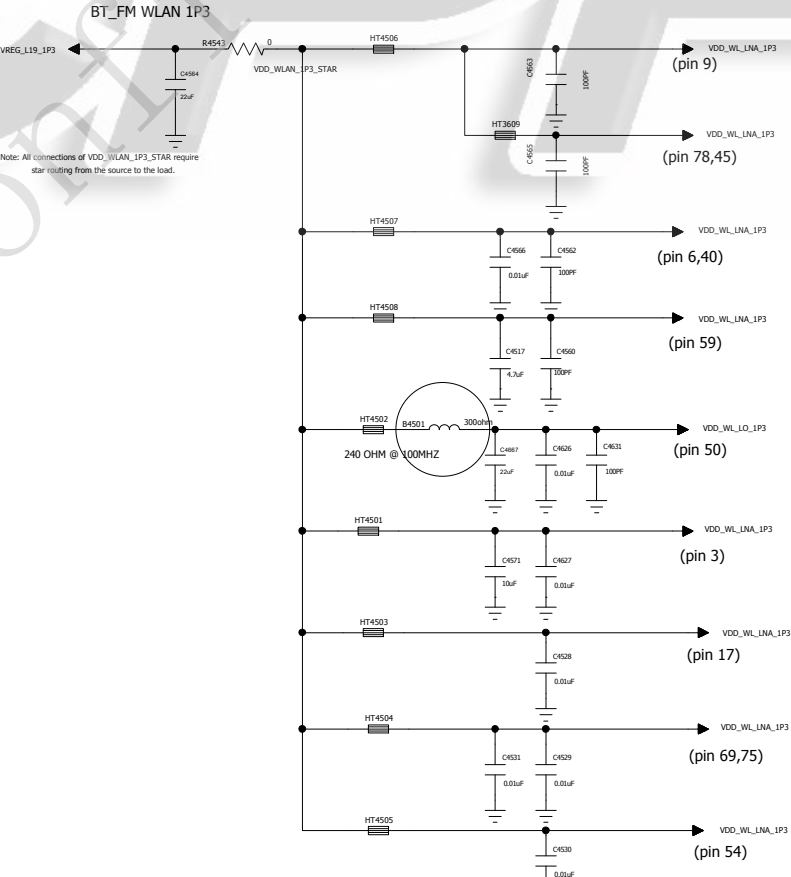
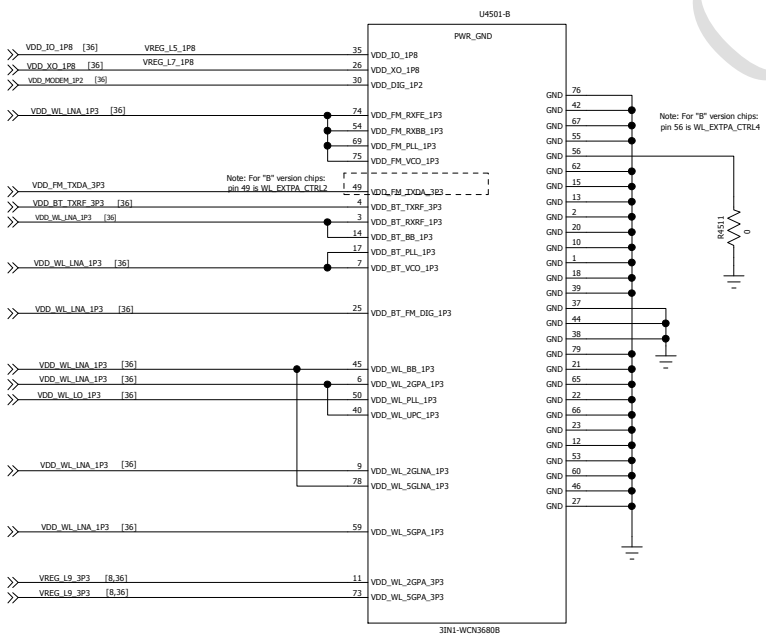
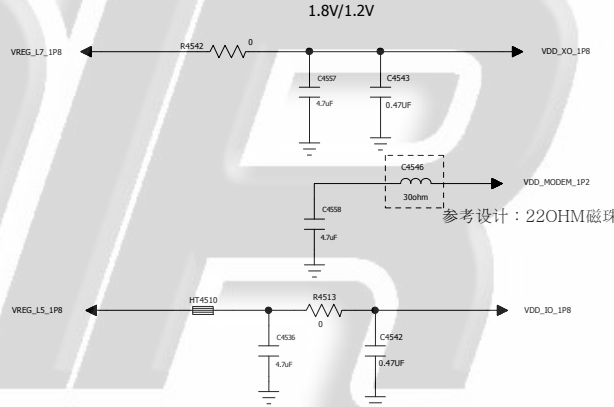
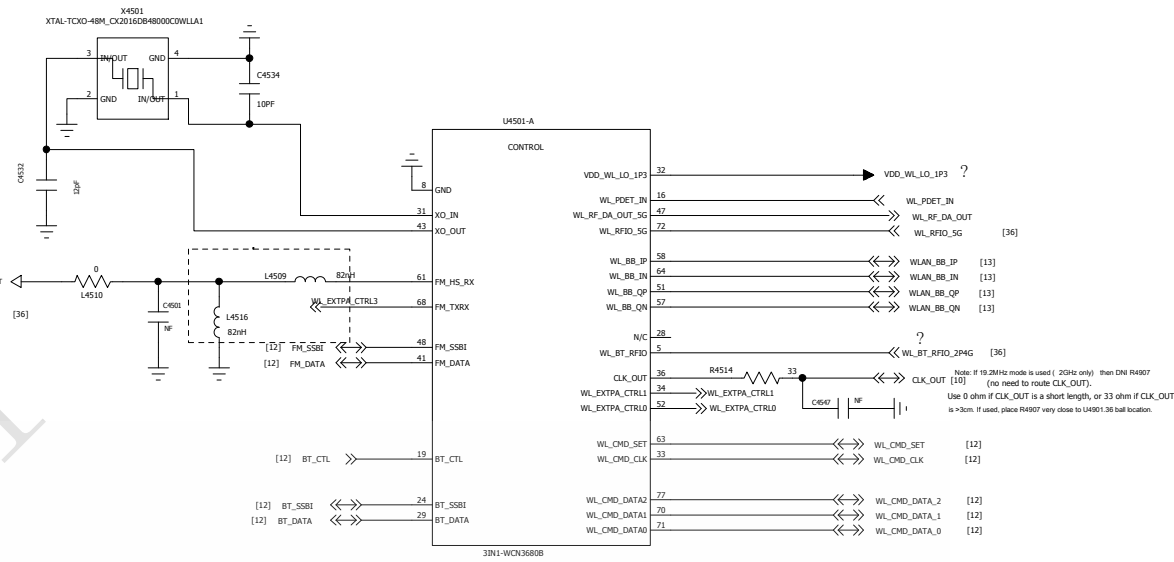


GPS



WIFI/BT

48 MHZ Crystal
Required for any design utilizing 5GHz



Note: If 2.4 GHz WLAN spurs are observed around 2.6 to 2.7 GHz, replace R4915 with ~20 nH inductor.